The Æther-flow Interpretation of Relativity

Primitive-Reservoir Observer-Reduction
Shell-Generating Substrate Lift Dynamics
Next Benchmark Criterion

Abstract

The current shell-restriction-frontier routing note had fixed the admissible burden order below the theorem forcing the split selector-shell core from primitive substrate shell restriction $\mathfrak{S}^{\mathrm{sub,sh}}$: first try to derive or uniquely force the shell-generating substrate lift package $\mathfrak{L}^{\mathrm{sub,lift}}$ from still more primitive explicit substrate structure, and if that could not yet be done cleanly, prove a genuinely smaller theorem-level collapse or sharper necessity / uniqueness result strictly inside shell restriction. The new lift-forcing theorem has now spent that first admissible route. The shell-generating substrate lift package $\mathfrak{L}^{\mathrm{sub,lift}}$ is no longer left as an independently inserted stopping point; it is a canonical and unique output of lift-generating substrate precursor dynamics $\mathfrak{P}^{\mathrm{sub,prec}}$.

This note records the routing consequence of that new theorem. The next honest benchmark-facing manuscript must therefore go one level below the current lift frontier: first derive or uniquely force the lift-generating substrate precursor package $\mathfrak{P}^{\mathrm{sub,prec}}$ from still more primitive explicit substrate structure in a way that actually changes benchmark-facing status, or, if that cannot be done cleanly, prove a genuinely smaller theorem-level collapse, sharper necessity result, or sharper uniqueness result strictly inside the shell-generating lift package $\mathfrak{L}^{\mathrm{sub,lift}}$. Another shell-restriction relay, split-core relay, selector-law restatement, combined-response restatement, ambient-package restatement, trace-core restatement, completion-core restatement, exact-shell affine nucleus restatement, or another lift relay that preserves the same $\mathfrak{L}^{\mathrm{sub,lift}}$ without shrinking or forcing it further does not count next.

The benchmark boundary remains fixed throughout. The one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, the low-energy matter law is not enlarged, and stronger first-principles promotion language remains paused. The positive result of this note is therefore routing discipline at the new lift-dynamics frontier.

1. Introduction

The active derivational program of Æther-flow Interpretation of Relativity is no longer at a stage where the shell-generating substrate lift package $\mathfrak{L}^{\mathrm{sub,lift}}$ should be treated as the default unexplained stopping point. The shell-restriction forcing theorem had already shown that the split zero-hidden selector-shell core $\mathfrak{C}^{\mathrm{sel,split}}$ is a canonical and unique output of primitive substrate shell restriction $\mathfrak{S}^{\mathrm{sub,sh}}$. The new lift-forcing theorem has now spent the next admissible upstream move below that frontier: $\mathfrak{L}^{\mathrm{sub,lift}}$ itself is a canonical and unique output of the deeper precursor class $\mathfrak{P}^{\mathrm{sub,prec}}$.

The present note records the routing consequence of that theorem. It does not reopen the lift mathematics, and it does not claim a completed first-principles substrate derivation of gravity. It

records only which kind of next manuscript would now count as an honest benchmark-facing gain below the current lift frontier.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays strictly inside that boundary. It records only which kind of next manuscript would now count as an honest benchmark-facing gain below the lift frontier.

2. Fixed Inputs

Definition 1 (Lift-frontier fixed inputs)**.** *The criterion recorded here uses the following fixed inputs.*

1. *The benchmark exact-closure package, which fixes the public one-metric GR target of the repository.*
2. *The benchmark gatekeeping layer, including the post-bridge status correction and the upstream-compression safeguard that blocks benchmark-neutral deeper-origin relays from counting as new front-line progress once the benchmark-relevant object is unchanged.*
3. *The shell-restriction forcing theorem, which proves that the split selector-shell core* $\mathfrak{C}^{\mathrm{sel,split}}$ *is a canonical and unique output of primitive substrate shell restriction* $\mathfrak{S}^{\mathrm{sub,sh}}$*.*
4. *The preserved lift-origin theorem, which already shows that the shell-generating substrate lift package* $\mathfrak{L}^{\mathrm{sub,lift}}$ *can be recovered canonically from lift-generating substrate precursor dynamics* $\mathfrak{P}^{\mathrm{sub,prec}}$*.*
5. *The new lift necessity / uniqueness theorem, which proves that* $\mathfrak{L}^{\mathrm{sub,lift}}$ *itself is a canonical and unique output of* $\mathfrak{P}^{\mathrm{sub,prec}}$*.*

Remark 1. *The present note does not replace those manuscripts. It records the routing consequence of reading them together under the active safeguard against recursive depth drift.*

3. Why the Lift Theorem Is the Frontier

Definition 2 (Benchmark-neutral lift relay)**.** *A* benchmark-neutral lift relay *is a manuscript that preserves the same benchmark one-metric output, the same ordinary matter route, the same shell-generating lift package* $\mathfrak{L}^{\mathrm{sub,lift}}$*, the same shell-restriction package* $\mathfrak{S}^{\mathrm{sub,sh}}$*, the same split selector-shell core* $\mathfrak{C}^{\mathrm{sel,split}}$*, the same selector-law consequence* $\mathfrak{S}^{\mathrm{sel},0}$*, the same combined-response consequence* $\mathfrak{R}^{\mathrm{comb}}$*, the same ambient-package consequence* $\mathfrak{A}^{\mathrm{bal}}$*, the same trace-core consequence* $\mathfrak{C}^{\mathrm{tr}}$*, the same completion-core consequence* $\mathfrak{C}^{\mathrm{zr,aff}}$*, the same exact-shell affine nucleus consequence* $\mathfrak{N}^{\mathrm{aff}}$*, and the*

same claim boundary while merely restating $\mathfrak{L}^{\mathrm{sub,lift}}$, $\mathfrak{S}^{\mathrm{sub,sh}}$, $\mathfrak{C}^{\mathrm{sel,split}}$, $\mathfrak{S}^{\mathrm{sel,0}}$, $\mathfrak{R}^{\mathrm{comb}}$, $\mathfrak{A}^{\mathrm{bal}}$, $\mathfrak{C}^{\mathrm{tr}}$, $\mathfrak{C}^{\mathrm{zr,aff}}$, $\mathfrak{N}^{\mathrm{aff}}$, *or the same precursor consequence without shrinking or forcing the lift package further.*

Proposition 1 (The lift-forcing theorem is the live frontier)**.** *At the current stage of the primitive-reservoir observer-reduction line, the theorem forcing the shell-generating substrate lift package* $\mathfrak{L}^{\mathrm{sub,lift}}$ *from lift-generating substrate precursor dynamics* $\mathfrak{P}^{\mathrm{sub,prec}}$ *is the live benchmark-facing frontier.*

Proof. That theorem changes benchmark-facing status at current scope because it removes the lift package $\mathfrak{L}^{\mathrm{sub,lift}}$ itself as the live internal stopping point. The current bridge datum is no longer merely a deeper package whose shell-restriction consequence was already known; it is now forced from the precursor layer. By contrast, a manuscript that preserves the same benchmark package, the same lift consequence, and the same shell-restriction/split-core consequences while merely restating the lift package or another already-forced larger or smaller bridge object does not alter benchmark-facing status unless it adds a comparably sharp gain. Therefore the live frontier is the lift-forcing theorem rather than the earlier shell-restriction forcing theorem or a later same lift relay. □

Corollary 1 (Status of the lift package as a next-step target)**.** *Under the current routing safeguard, the shell-generating substrate lift package* $\mathfrak{L}^{\mathrm{sub,lift}}$ *is no longer itself the default next benchmark-facing target.*

Proof. The lift-forcing theorem already proves that $\mathfrak{L}^{\mathrm{sub,lift}}$ is a canonical and unique output of $\mathfrak{P}^{\mathrm{sub,prec}}$. Once that is on record, another manuscript that spends the move on the lift package again does not remove any further benchmark-relevant freedom. The open burden has therefore moved below lift scope. □

4. The Next Benchmark Criterion

Definition 3 (Admissible next benchmark-facing lift manuscript)**.** *At the current lift frontier, a manuscript counts as an admissible next benchmark-facing move only if it respects the following burden order:*

1. *its primary attempt is to derive or uniquely force the lift-generating substrate precursor package* $\mathfrak{P}^{\mathrm{sub,prec}}$ *from still more primitive explicit substrate structure in a way that does more than merely relocate the unexplained layer deeper;*

2. *if that cannot be done cleanly at current scope, its admissible fallback gain is to prove a genuinely smaller theorem-level collapse, sharper necessity result, or sharper uniqueness result strictly inside the shell-generating lift package* $\mathfrak{L}^{\mathrm{sub,lift}}$*;*

3. *in either case it preserves the same benchmark one-metric package, the same ordinary matter route, and the same claim boundary, and it introduces neither a second operative metric nor an enlarged low-energy matter law and licenses no stronger first-principles promotion language.*

Theorem 1 (Next honest benchmark-facing task at lift scope)**.** *The next honest benchmark-facing manuscript at the current lift frontier should therefore go one level below that frontier: first try to derive or uniquely force the lift-generating substrate precursor package* $\mathfrak{P}^{\mathrm{sub,prec}}$ *from still more primitive explicit substrate structure in a way that actually changes benchmark-facing status. If*

that cannot be done cleanly, the admissible fallback is to prove a genuinely smaller collapse, sharper necessity result, or sharper uniqueness result strictly inside the shell-generating lift package $\mathfrak{L}^{\mathrm{sub,lift}}$*. A manuscript that only restates the shell-restriction package* $\mathfrak{S}^{\mathrm{sub,sh}}$*, only restates the split selector-shell core* $\mathfrak{C}^{\mathrm{sel,split}}$*, only restates the selector law* $\mathfrak{S}^{\mathrm{sel},0}$*, only restates the combined response law* $\mathfrak{R}^{\mathrm{comb}}$*, only restates the ambient package* $\mathfrak{A}^{\mathrm{bal}}$*, only restates the trace core* $\mathfrak{C}^{\mathrm{tr}}$*, only restates the completion-core presentation* $\mathfrak{C}^{\mathrm{zr,aff}}$*, only restates the exact-shell affine nucleus* $\mathfrak{N}^{\mathrm{aff}}$*, or only repeats the same lift consequence while preserving* $\mathfrak{L}^{\mathrm{sub,lift}}$ *without shrinking or forcing it further does not satisfy the criterion.*

Proof. By Proposition 1, the live frontier already lies at the theorem that forces the lift package from precursor dynamics. Therefore a second manuscript below it must improve on that status rather than merely accompany it. A deeper-origin manuscript can do so only if it changes benchmark-facing status by more than depth alone, and the first clean way to do that is to force $\mathfrak{P}^{\mathrm{sub,prec}}$ itself from still more primitive explicit substrate structure. If that first option is not yet available cleanly, the routing safeguard allows the fallback of a smaller internal collapse, necessity result, or uniqueness result inside $\mathfrak{L}^{\mathrm{sub,lift}}$. If a manuscript simply reproduces the same lift consequence through another relay while preserving the same lift package without shrinking or forcing it further, the routing rule classifies it as benchmark neutral. □

5. What the Next Move Should Not Be

Proposition 2 (Screened next moves)**.** *At the current lift frontier, none of the following is the default next benchmark-facing move:*

1. *another restatement of the primitive substrate shell-restriction package* $\mathfrak{S}^{\mathrm{sub,sh}}$*;*
2. *another restatement of the split zero-hidden selector-shell core* $\mathfrak{C}^{\mathrm{sel,split}}$*;*
3. *another restatement of the full zero-hidden selector law* $\mathfrak{S}^{\mathrm{sel},0}$*;*
4. *another restatement of the combined response substrate law* $\mathfrak{R}^{\mathrm{comb}}$*;*
5. *another restatement of the completion-balance ambient dynamics package* $\mathfrak{A}^{\mathrm{bal}}$*;*
6. *another restatement of the completion-state shell-trace core* $\mathfrak{C}^{\mathrm{tr}}$*;*
7. *another restatement of the full completion-core presentation* $\mathfrak{C}^{\mathrm{zr,aff}}$*;*
8. *another restatement of the exact-shell affine nucleus* $\mathfrak{N}^{\mathrm{aff}}$*;*
9. *another lift relay that preserves the same* $\mathfrak{L}^{\mathrm{sub,lift}}$ *without shrinking or forcing it further;*
10. *a manuscript that introduces a second operative metric, enlarges the ordinary matter law, or uses stronger first-principles promotion language.*

Proof. Items (1)–(9) fail the preceding criterion because they do not directly address the current precursor-facing burden. They revisit either already-forced larger bridge objects, already-forced smaller bridge objects, or the same lift stopping point rather than removing the remaining lift freedom or proving a still smaller internal datum inside $\mathfrak{L}^{\mathrm{sub,lift}}$. Item (10) violates the fixed claim boundary of the benchmark package and therefore cannot count as a disciplined next step on the active line. The benchmark package remains the exact one-metric GR package already fixed by

adoption, so any admissible next manuscript must preserve that boundary while keeping stronger first-principles promotion language paused. □

Remark 2. *This screening statement is not a denial that the shell-restriction forcing theorem, the selector-law collapse theorem, the selector-law forcing theorem, the combined-response theorem, the ambient-package theorem, the trace-core theorem, or the completion-core theorem matter historically. It is a routing statement: they are not the honest way to spend the next benchmark-facing move from the current lift frontier.*

6. Conclusion

The positive result of this note is routing discipline at the current lift frontier. The lift-forcing theorem remains the live benchmark-facing gain because it already removes the shell-generating substrate lift package as the internal stopping point on the active line. The lift package therefore no longer sets the honest next burden, and neither do the already-forced shell-restriction package, split core, selector law, combined response law, ambient package, trace core, completion-core presentation, or exact-shell affine nucleus.

The scope remains narrow and honest. The benchmark package is unchanged: the one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged. Nothing here upgrades adoption to derivation or licenses stronger first-principles promotion language yet.

The next open burden is therefore clear. The next benchmark-facing manuscript should go one level below the current lift frontier: first try to derive or uniquely force the lift-generating substrate precursor package $\mathfrak{P}^{\mathrm{sub,prec}}$ from still more primitive explicit substrate structure in a way that actually changes benchmark-facing status while preserving the same benchmark one-metric package and claim boundary. If that cannot be done cleanly, the admissible fallback is a genuinely smaller collapse, sharper necessity result, or sharper uniqueness result strictly inside the shell-generating lift package $\mathfrak{L}^{\mathrm{sub,lift}}$. Another shell-restriction relay, split-core relay, selector-law restatement, combined-response restatement, ambient-package restatement, trace-core restatement, completion-core restatement, exact-shell affine nucleus restatement, or another lift relay that preserves the same lift package without shrinking or forcing it further is not the clean next manuscript target at current scope.